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2.1.4

$$(P_1 \Rightarrow P_2) \Rightarrow ((P_2 \vee p) \Rightarrow (P_1 \vee p))$$

$$P_1 \Rightarrow P_2 = P_2 \vee \overline{P_1}$$

$$(P_1 \vee p) \Rightarrow (p \vee P_2) = p \vee P_2 \vee \overline{P_1}$$

$$(P_1 \Rightarrow P_2) \Rightarrow ((P_1 \vee p) \Rightarrow (P_2 \vee p)) = 1$$

p	P_1	P_2	R
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

DNSh = KNSh = MDNSh = MKNSh = 1

2.1.6

$$((p = q) \wedge (q = r)) \Rightarrow (p \Rightarrow r)$$

$$p \Rightarrow q = q \vee \overline{p}$$

$$q = r = r \vee \overline{q}$$

$$(p \Rightarrow q) \wedge (q \Rightarrow r) = (q \wedge r) \vee (\overline{p} \wedge \overline{q})$$

$$p \Rightarrow r = r \vee \overline{p}$$

$$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r) = 1$$

p	q	r	R
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

DNSh = KNSh = MDNSh = MKNSh = 1

2.1.11

$\bar{P}_1 \Rightarrow \neg(P_1 \Rightarrow P_2)$

$P_1 \Rightarrow P_2 = P_2 \vee \bar{P}_1$

$P_1 \not\Leftarrow P_2 = P_1 \wedge \bar{P}_2$

$\bar{P}_1 \Rightarrow P_1 \Rightarrow P_2 = P_1$

P_1	P_2	R
0	0	0
0	1	0
1	0	1
1	1	1

DNSH = KNSH = MDNSH = MKNSH = P_1

2.1.16

$(x \Rightarrow y) \wedge (y \Rightarrow z) \Rightarrow (x \Rightarrow z)$

$x \Rightarrow y = y \vee \bar{x}$

$y \Rightarrow z = z \vee \bar{y}$

$(x \Rightarrow y) \wedge (y \Rightarrow z) = (y \wedge z) \vee (\bar{x} \wedge \bar{y})$

$z \Rightarrow x = x \vee \bar{z}$

$((x \Rightarrow y) \wedge (y \Rightarrow z)) \Rightarrow (z \Rightarrow x) = x \vee \bar{z}$

x	y	z	R
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

MONSH = MKNSh = DNSH = KNSH = $\underline{x \vee \bar{z}}$

2.1.21 $((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$

$$p \Rightarrow q = q = \bar{p}$$

$$q \Rightarrow r = r \vee \bar{q}$$

$$(p \Rightarrow q) \wedge (q \Rightarrow r) = (q \wedge r) \vee (\bar{p} \wedge \bar{q})$$

$$p \Rightarrow r = r \vee \bar{p}$$

$$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r) = 1$$

p	q	r	R
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

DNSH = KNSH = MDNSh = MKNSH = 1

2.1.29

$$\overline{((x \Rightarrow z) \Rightarrow ((y \Rightarrow z) \Rightarrow (x \vee y \Rightarrow z)))}$$

$$x \Rightarrow z = z \vee \bar{x}$$

$$y \Rightarrow z = z \vee \bar{y}$$

$$(x \vee y) \Rightarrow z = z \vee (\bar{x} \wedge \bar{y})$$

$$(y \Rightarrow z) \Rightarrow ((x \vee y) \Rightarrow z) = y \vee z \vee \bar{x}$$

$$(x \Rightarrow z) \Rightarrow ((y \Rightarrow z) \Rightarrow ((x \vee y) \Rightarrow z)) = 1$$

$$(x \Rightarrow z) \not\Rightarrow ((y \Rightarrow z) \Rightarrow ((x \vee y) \Rightarrow z)) = 0$$

x	y	z	R
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

DNSH = KNSH = MDNSH = MKNSH = 0

2.1.25

$(x \wedge \neg(x \wedge \bar{x} \Rightarrow y \wedge \bar{y}) \Rightarrow z) \vee x \vee (y \wedge z) \vee (y \wedge z)$

$x \wedge \bar{x} = 0$

$y \wedge \bar{y} = 0$

$(x \wedge \bar{x}) \Rightarrow (y \wedge \bar{y}) = 1$

$(x \wedge \bar{x}) \not\Rightarrow (y \wedge \bar{y}) = 0$

$x \wedge (x \wedge \bar{x}) \not\Rightarrow (y \wedge \bar{y}) = 0$

$(x \wedge (x \wedge \bar{x}) \not\Rightarrow (y \wedge \bar{y})) \Rightarrow z = 1$

$x \vee (y \wedge z) \vee ((x \wedge (x \wedge \bar{x}) \not\Rightarrow (y \wedge \bar{y})) \Rightarrow z) = 1$

x	y	z	R
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

DNSH = KNSH = MDNSH = MKNSH = 1

21.19

$$(x \vee \bar{y} \Rightarrow (z \Rightarrow y \vee \bar{y} \vee x)) \wedge (x \vee \overline{x \Rightarrow (x \Rightarrow x)}) = y$$

$$x \Rightarrow x = 1$$

$$x \Rightarrow (x \Rightarrow x) = 1$$

$$x \not\Rightarrow (x \Rightarrow x) = 0$$

$$x \vee x \not\Rightarrow (x \not\Rightarrow x) = x$$

$$x \vee y \vee \bar{y} = 1$$

$$x \Rightarrow (x \vee y \vee \bar{y}) = 1$$

$$(x \vee \bar{y}) \Rightarrow (z \Rightarrow (x \vee y \vee \bar{y}))) \wedge (x \vee x \not\Rightarrow (x \Rightarrow x)) =$$

$$((x \vee \bar{y}) \Rightarrow (z \Rightarrow (x \vee y \vee \bar{y}))) \wedge (x \vee x \not\Rightarrow (x \Rightarrow x))) \Rightarrow y =$$

$$= y \vee \bar{x}$$

x	y	z	R
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

DNSH = KNSH = MDNSH = MKNSH = $y \vee \bar{x}$

2.1.22

$$\neg((P_1 \Rightarrow P_2) \Rightarrow ((P_1 \wedge p) \Rightarrow (P_2 \wedge p)))$$

$$P_1 \Rightarrow P_2 = P_2 \vee \overline{P_1}$$

$$(p \wedge P_1) \Rightarrow (p \wedge P_2) = P_2 \vee \overline{p} \vee \overline{P_1}$$

$$(p \wedge P_1) \Rightarrow ((p \wedge P_1) \Rightarrow (p \wedge P_2)) = 1$$

$$(P_1 \Rightarrow P_2) \not\Rightarrow ((p \wedge P_1) \Rightarrow (p \wedge P_2)) = 0$$

P	P_1	P_2	R
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

DNSH = KNSH = MDNSH = MKNSH = 0

2.1 2

$((p \to q) \wedge (q \to r)) \to (p \to r)$

$p \Rightarrow q = q \vee \bar{p}$

$q \Rightarrow r = r \vee \bar{q}$

$(p \Rightarrow q) \wedge (q \Rightarrow r) = (q \wedge r) \vee (\bar{p} \wedge \bar{q})$

$p \Rightarrow r = r \vee \bar{p}$

$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r) = 1$

p	q	r	R
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

DNSH = KNSH = MDNSH = MKNSH = 1